DESIGN COMPUTED · FREE GUIDE

Round the corners: how a tower tames the wind

The free guide to the reel on 30 St Mary Axe (the Gherkin): why wind dives down a tall flat wall, how a rounded plan keeps the street calmer, and how to test wind on your own model.



How it works

Wind hits a tall flat wall and part of it is pushed down the face to the street, where pedestrians feel it as gusts. A rounded plan lets the wind flow around the tower, like water around a stone, so less is sent to the ground.

A round shape also reduces the wind load on the cladding and the structure. At 30 St Mary Axe the plan is circular, widens in profile as it rises, and tapers towards the top. The pressure differences around the building are also used by its natural ventilation.

Key facts

30 St Mary Axe, London, was designed by Foster + Partners with Arup. It is about 180 m high with 41 floors, completed in December 2003 and opened in 2004. Its structure is described as a trussed tube with diagonal bracing at the perimeter, which leaves the floors free of columns.

Wind-tunnel testing is standard for tall buildings. We did not find public results for the Gherkin's own tests, so this guide does not quote any.

Built examples

30 St Mary Axe, London (2004)	Foster + Partners with Arup: circular plan, less wind sent to the street.
Taipei 101, Taipei (2004)	Stepped profile and corner changes reduce vortex shedding; a 660-tonne damper does the rest.
Shanghai Tower, Shanghai (2015)	Taper and twist tuned with wind-tunnel and computer modelling.
Burj Khalifa, Dubai (2010)	Stepped setbacks and a rotated base improve its response to the strongest wind.

Try it yourself

1. Cut two tower models from card: one square plan, one round plan, each about 30 cm high.

2. Put a table fan at one end of a table and a thin strip of tissue next to the base of each tower.
3. Watch where the air goes: does it dive down the face, or flow around?
4. Film each test and compare the movement of the tissue at the base.
5. Try a third shape (rounded or chamfered corners) and compare.
6. For a digital test, try a free wind-flow tool such as the open-source OpenFOAM (advanced), or a simple online wind simulator.

The rule to remember

Next time you draw a tall block, round the corners and test the wind.

Testing late	Test the shape in early sketches, not at the end.
Ignoring the street	Wind at pedestrian level decides whether a place is used.
Copying a famous shape	Wind depends on the site: test your own.

Go further

30 St. Mary Axe, Designing Buildings
https://www.designingbuildings.co.uk/wiki/30_St._Mary_Axe

Form follows physics, Structure magazine
https://www.structuremag.org/article/form-follows-physics/

Skyscraper wind tunnel, Skyscraper Museum
https://old.skyscraper.org/EXHIBITIONS/TEN_TOPS/windtunnel.php

Credits
Photos: see the reel caption (Wikimedia Commons, free licences).
Facts: Designing Buildings, Structure magazine, Skyscraper Museum.

Made by Design Computed · Follow for more · @designcomputed